

Portes logiques

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1) La base des portes logiques :

1-1) NON (NOT) {inverseur} :

Américain



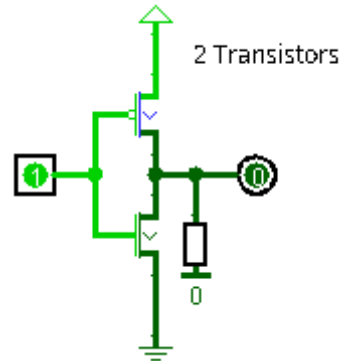
Europe



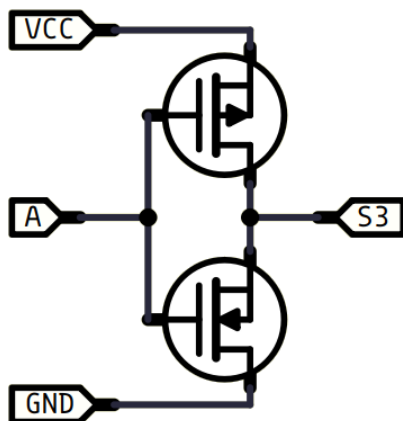
A	S
0	1
1	0

Transistors

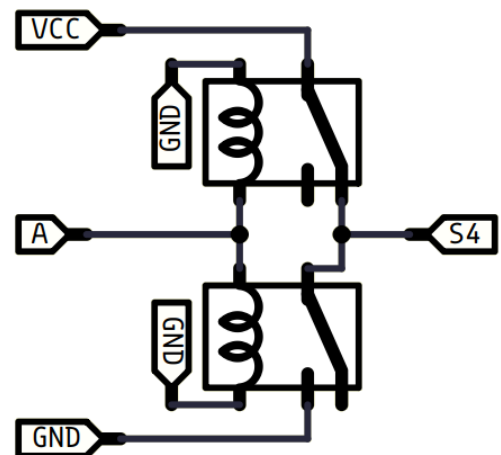
(logisim)



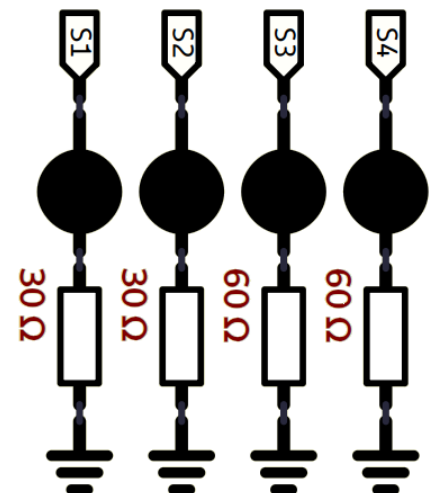
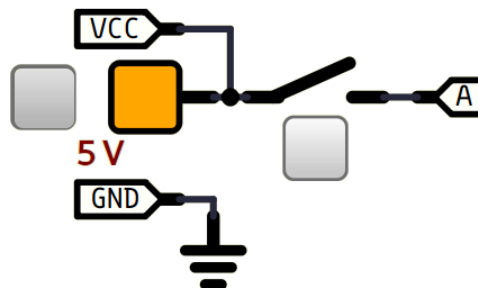
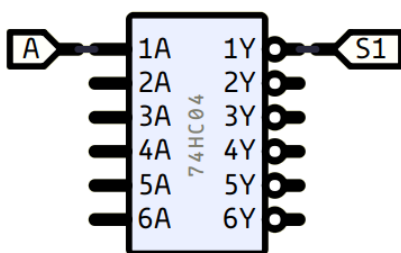
(simulide)

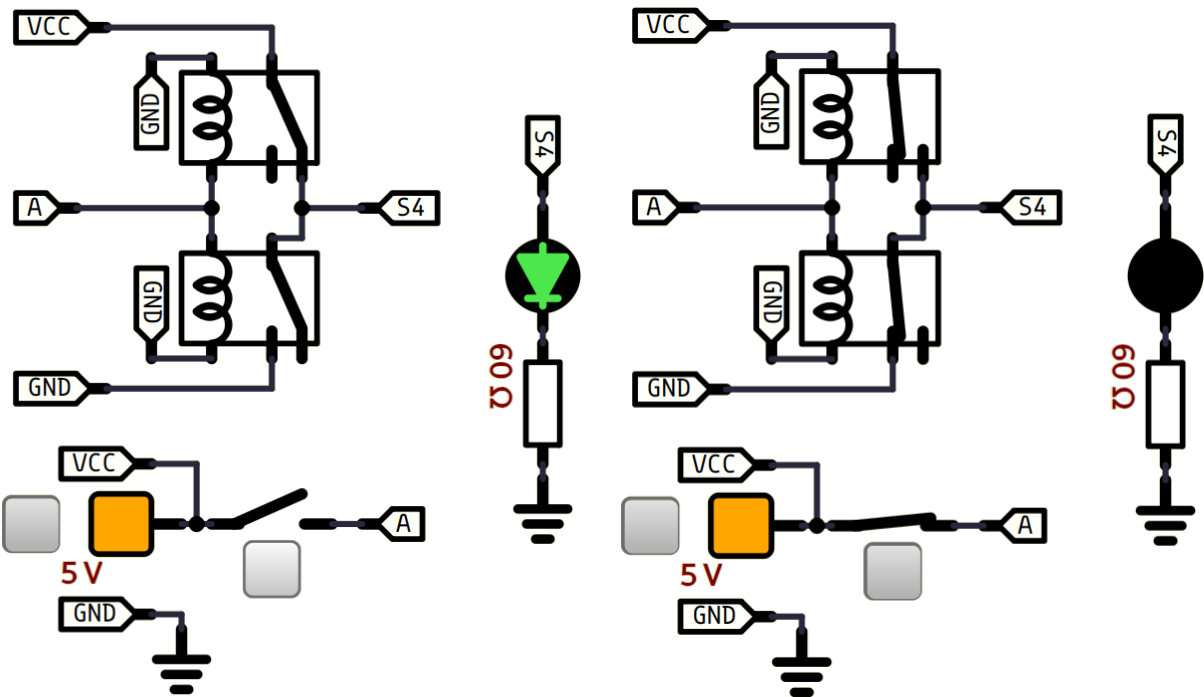


NOT



74HC04-46





1-2) ET (AND) :

Américain



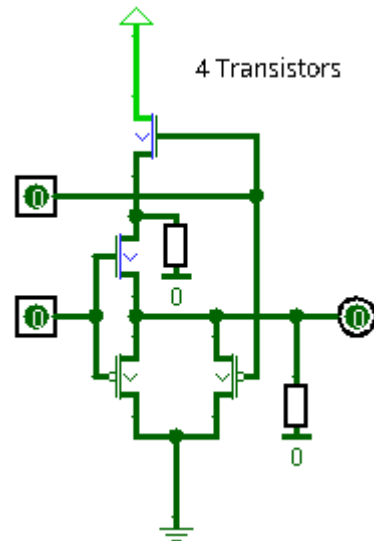
Europe



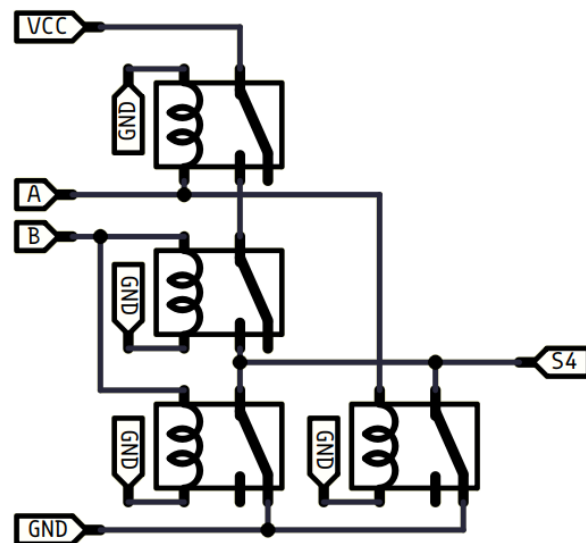
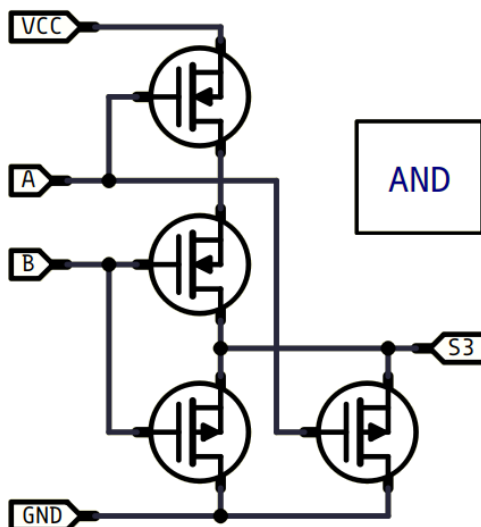
A	B	S
0	0	0
0	1	0
1	0	0
1	1	1

Transistors

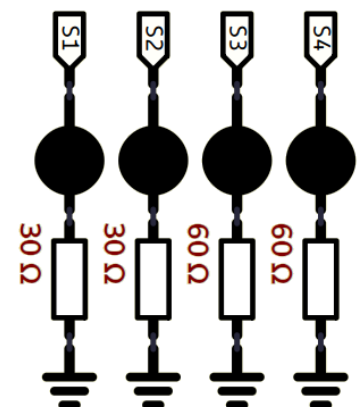
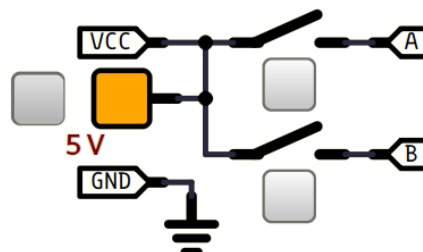
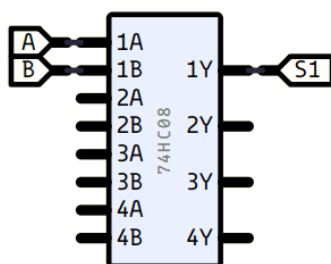
(logisim)

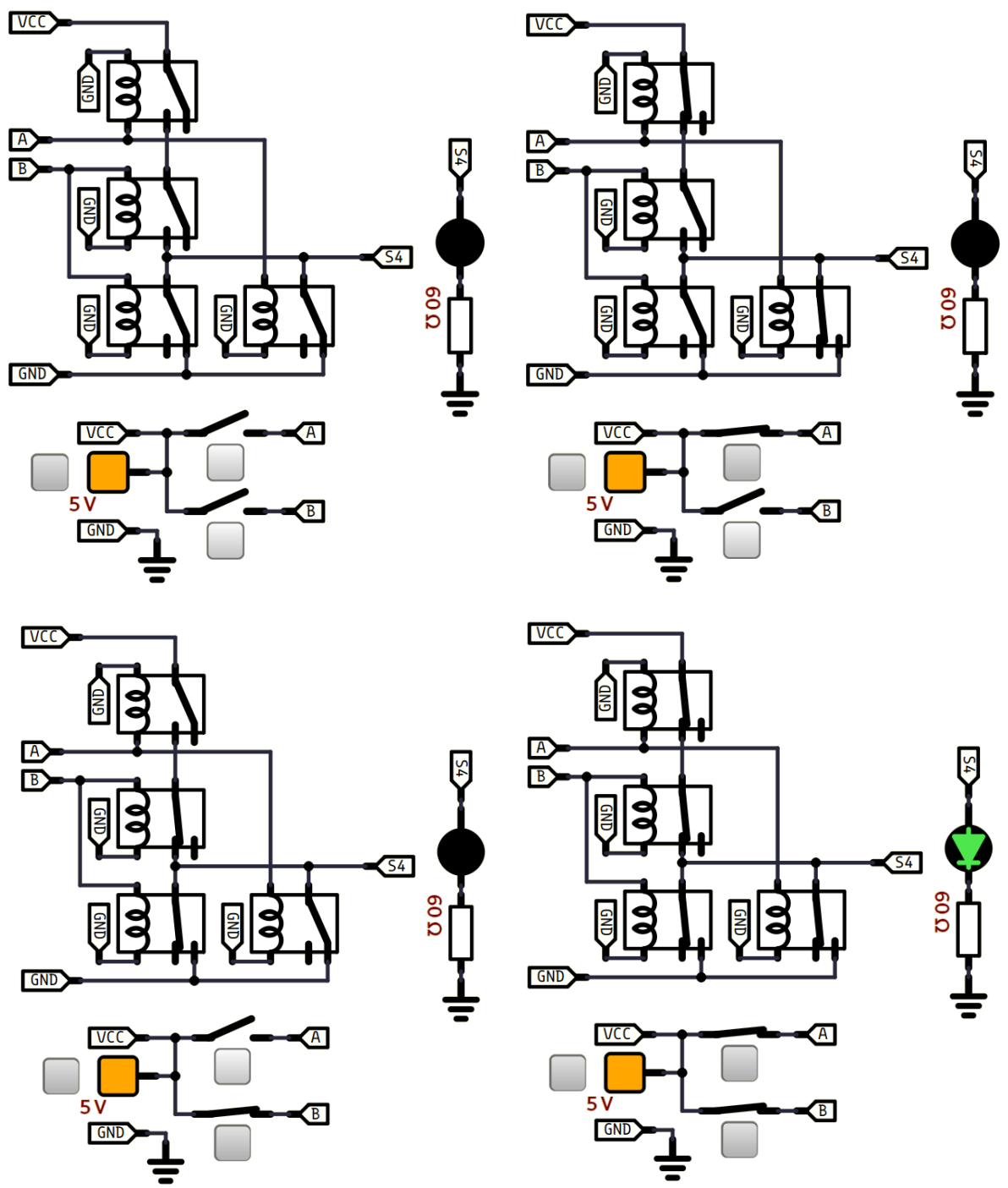


(simulide)



74HC08-279





1-3) NON-ET (NAND) :

Américain



Europe



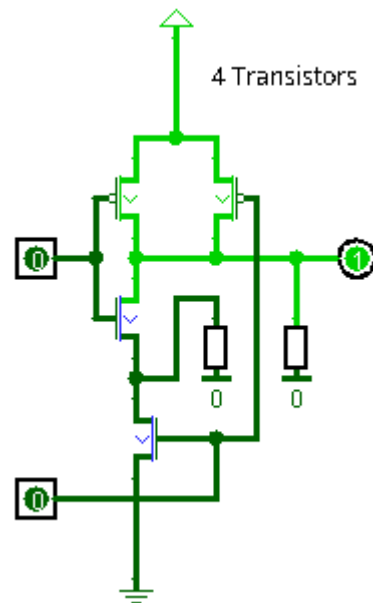
A	B	S
0	0	1
0	1	1
1	0	1
1	1	0

Équivalences

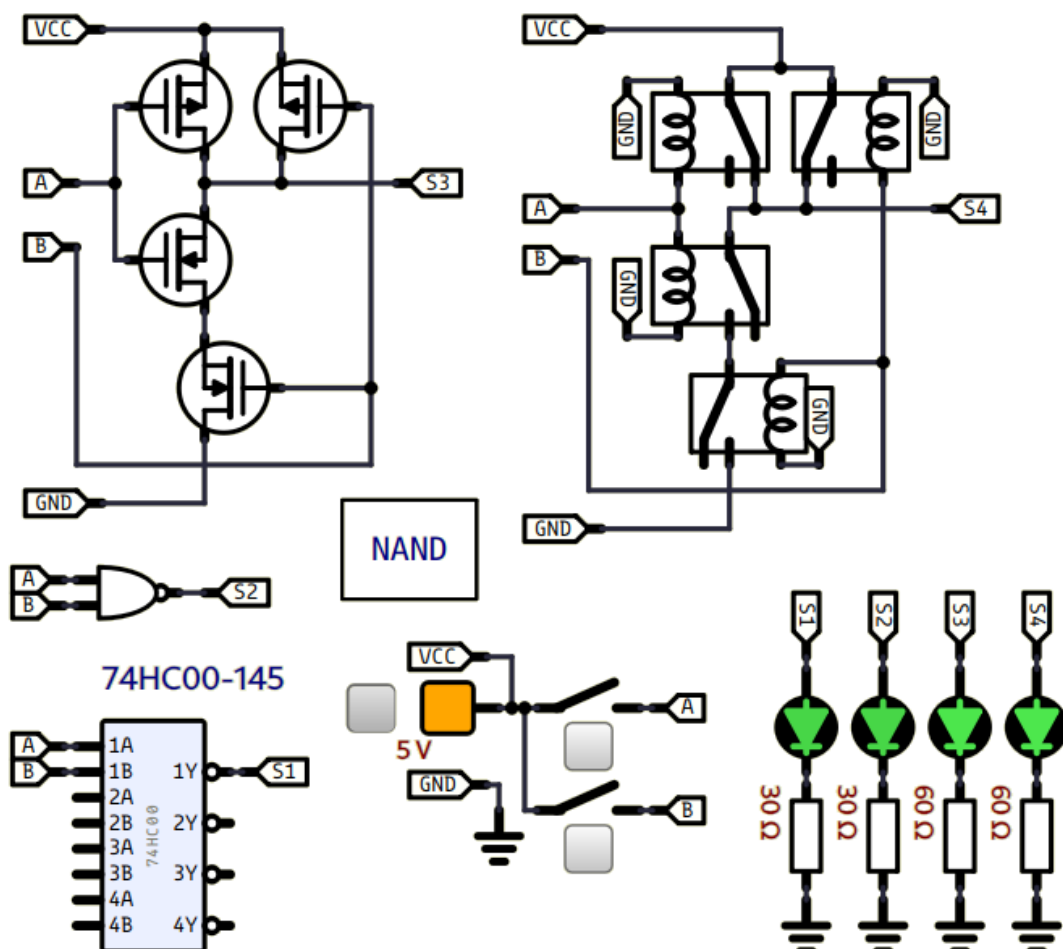


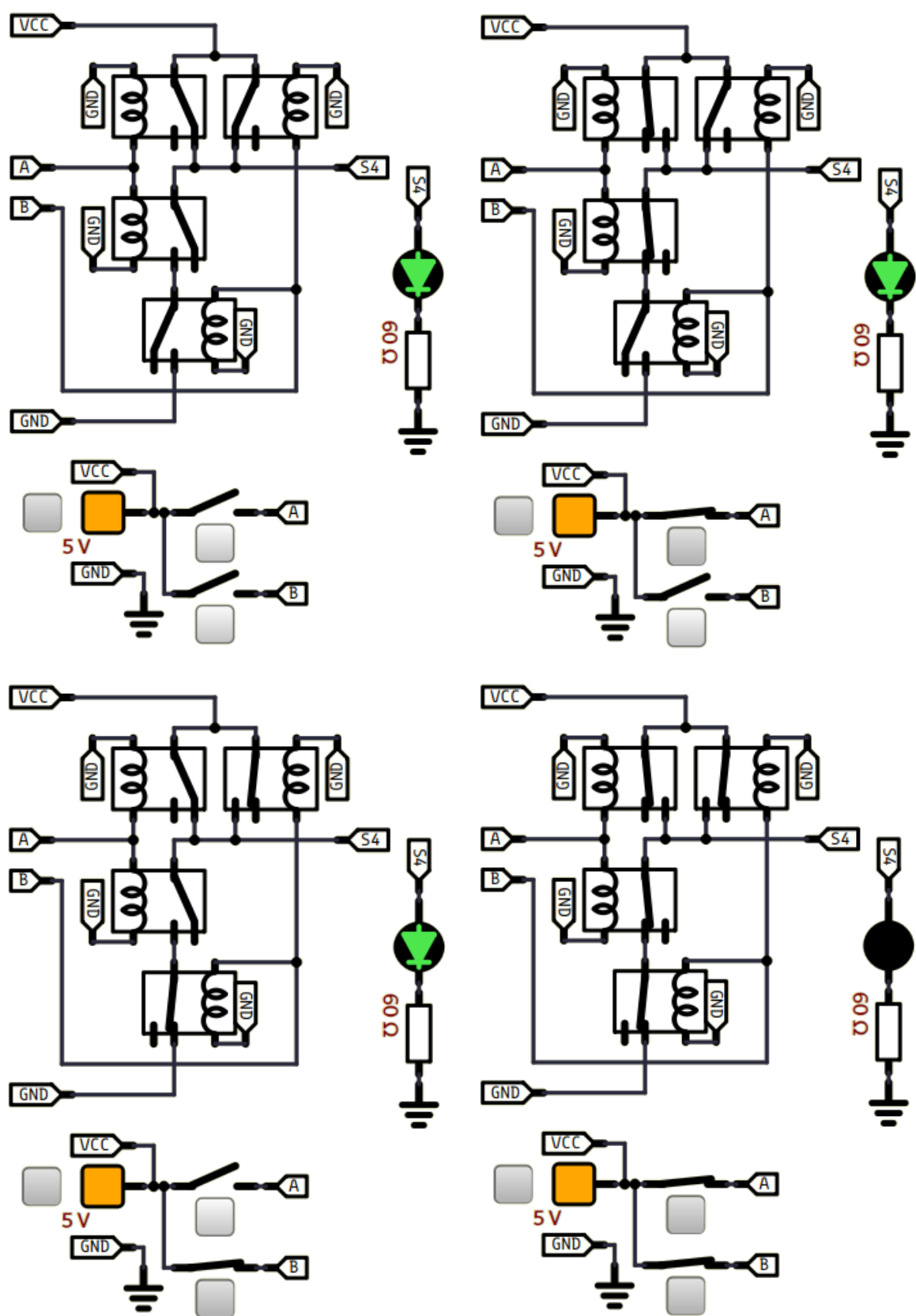
Transistors

(logisim)



(simulide)





1-4) OU (OR) :

Américain



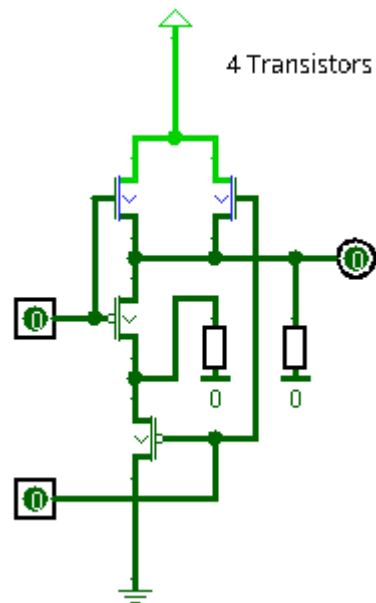
Europe



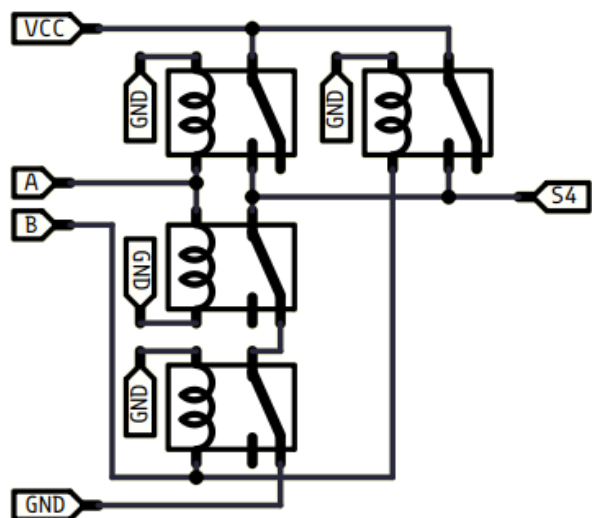
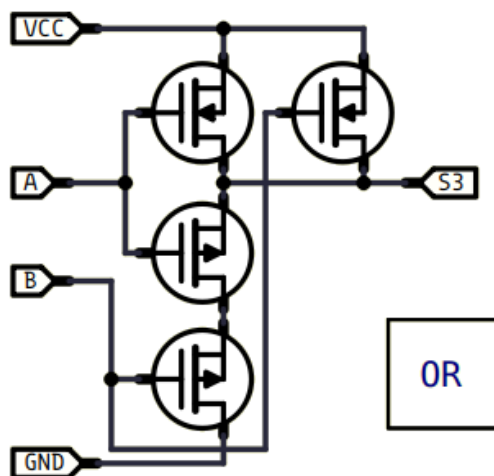
A	B	S
0	0	0
0	1	1
1	0	1
1	1	1

Transistors

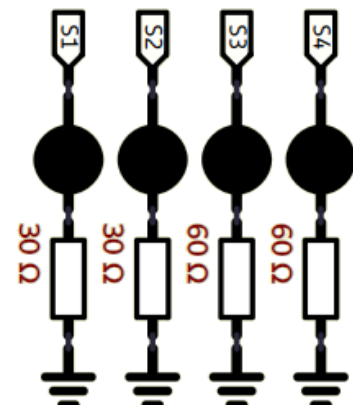
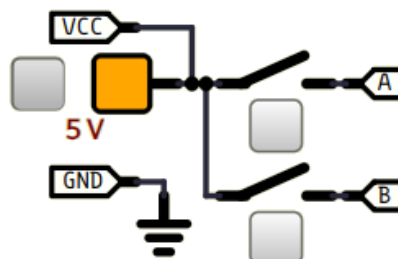
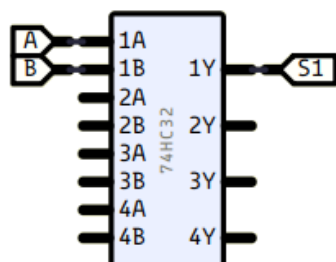
(logisim)

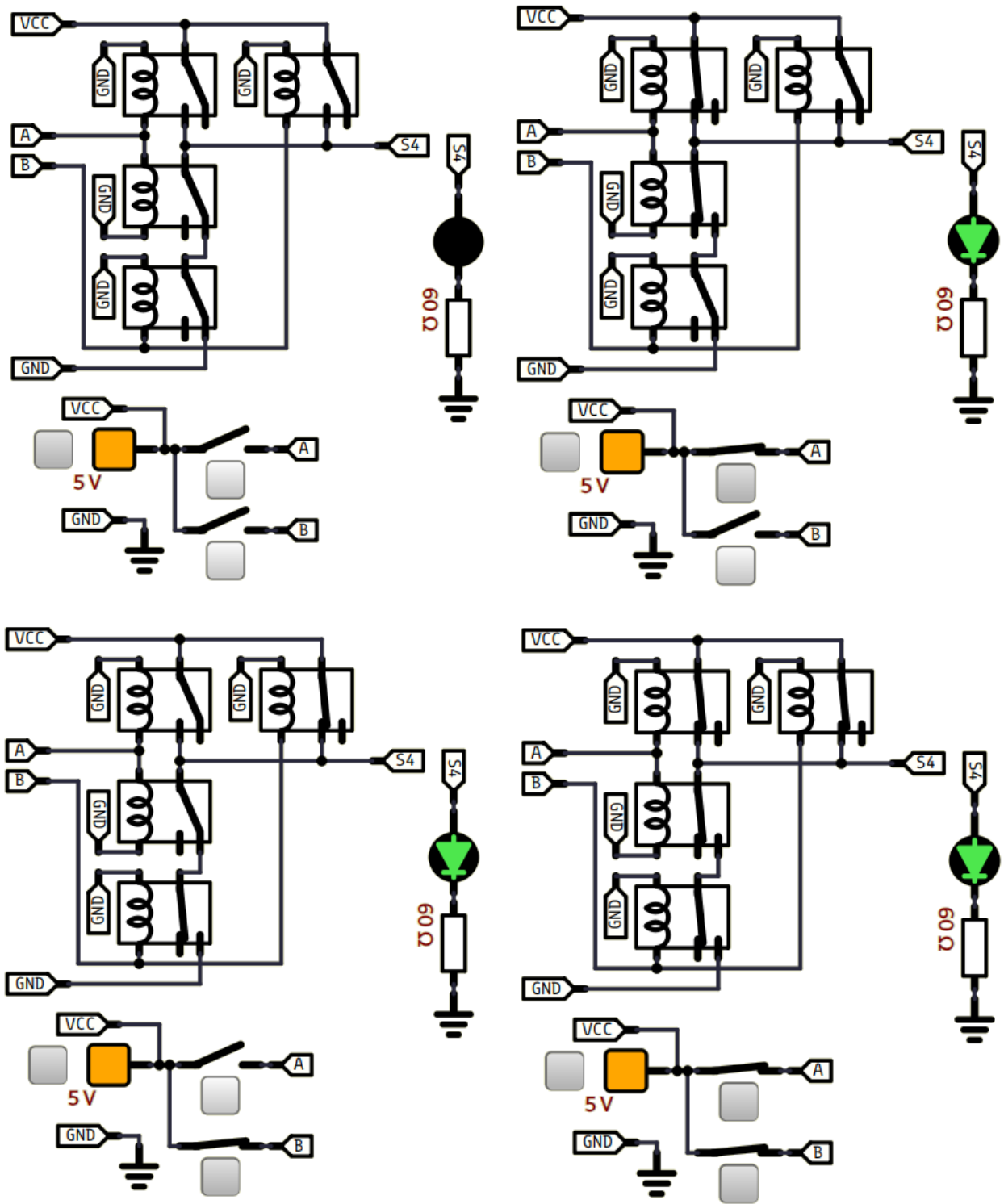


(simulide)



74HC32-100





1-5) NON-OU (NOR) :

Américain

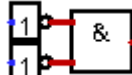
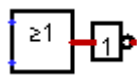


Europe



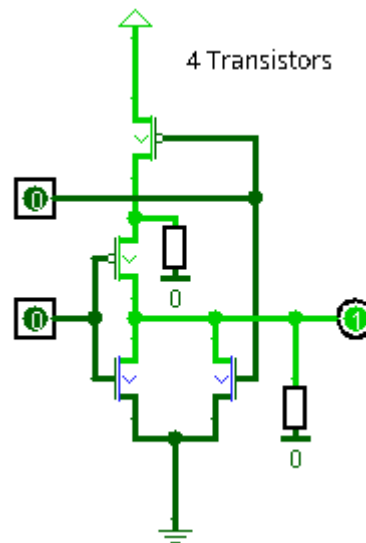
A	B	S
0	0	1
0	1	0
1	0	0
1	1	0

Équivalences

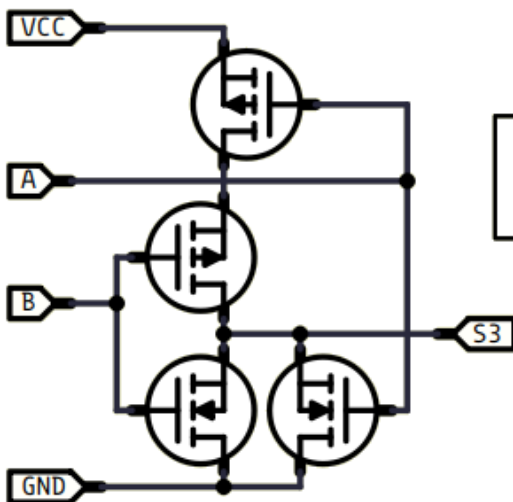


Transistors

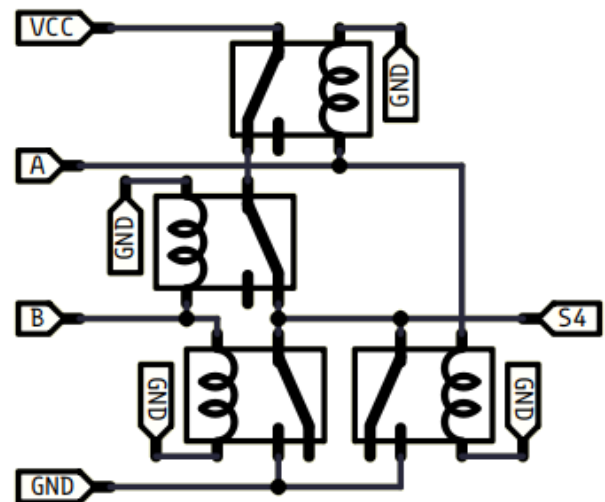
(logisim)



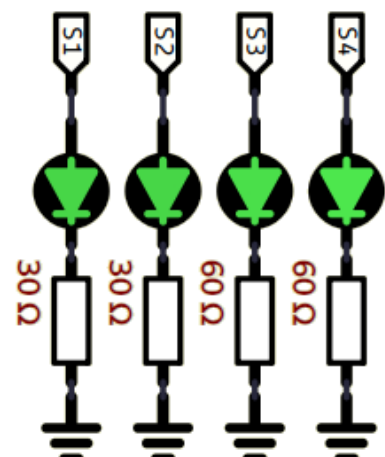
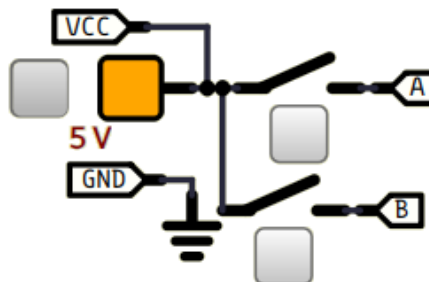
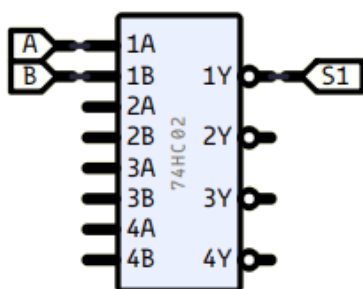
(simulide)

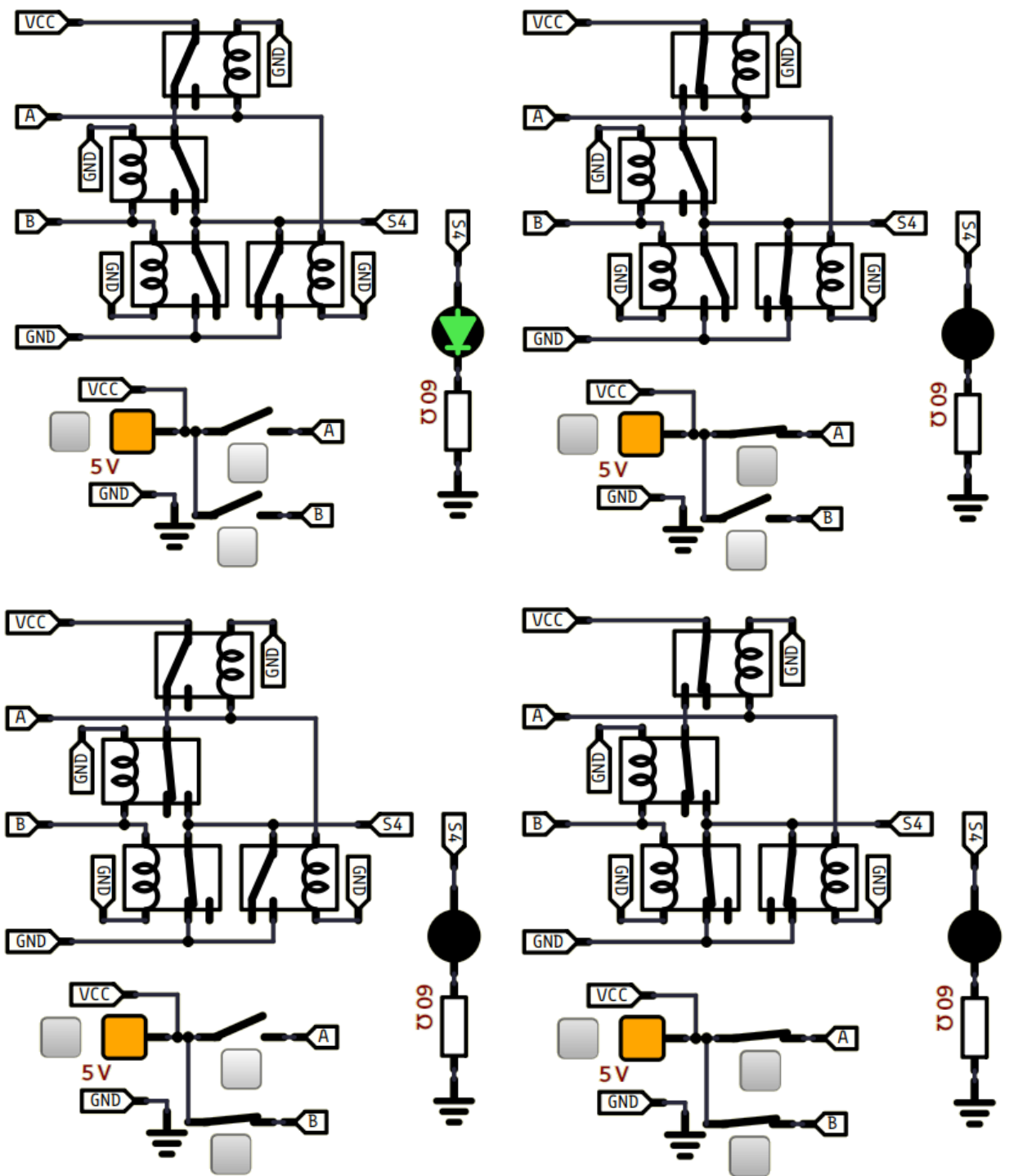


NOR



74HC02-90





1-6) OU-EXCLUSIF (XOR) :

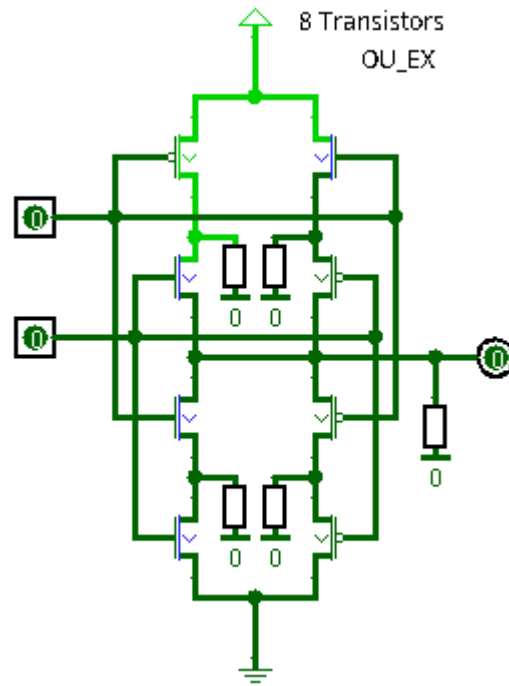
Américain



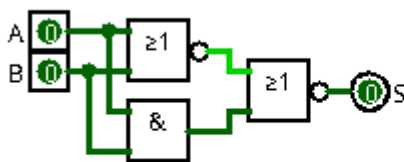
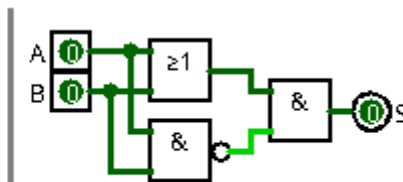
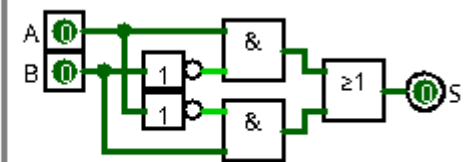
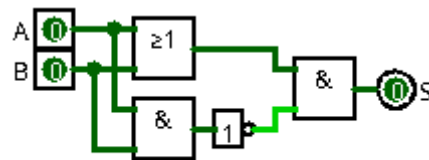
Europe



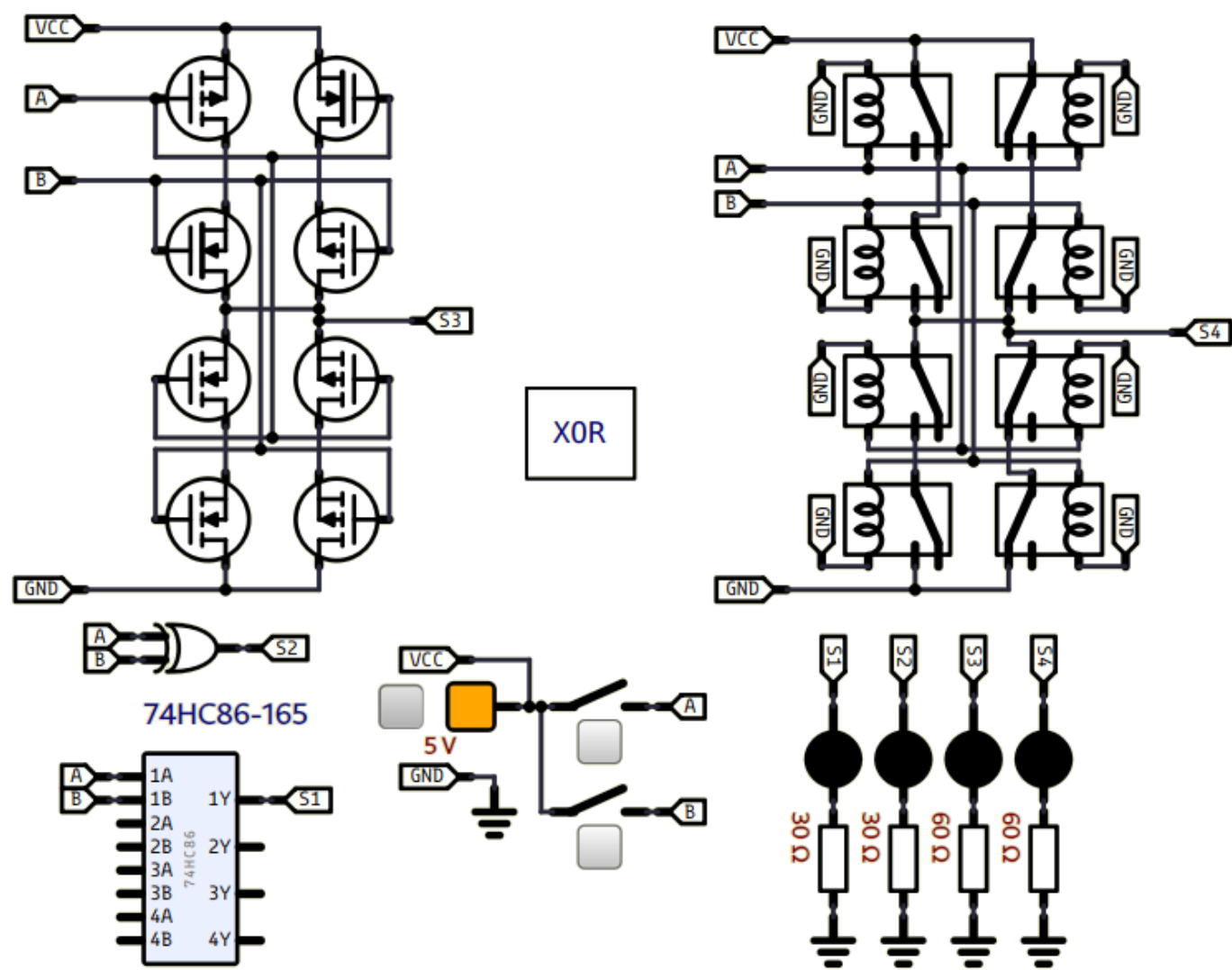
A	B	S
0	0	0
0	1	1
1	0	1
1	1	0

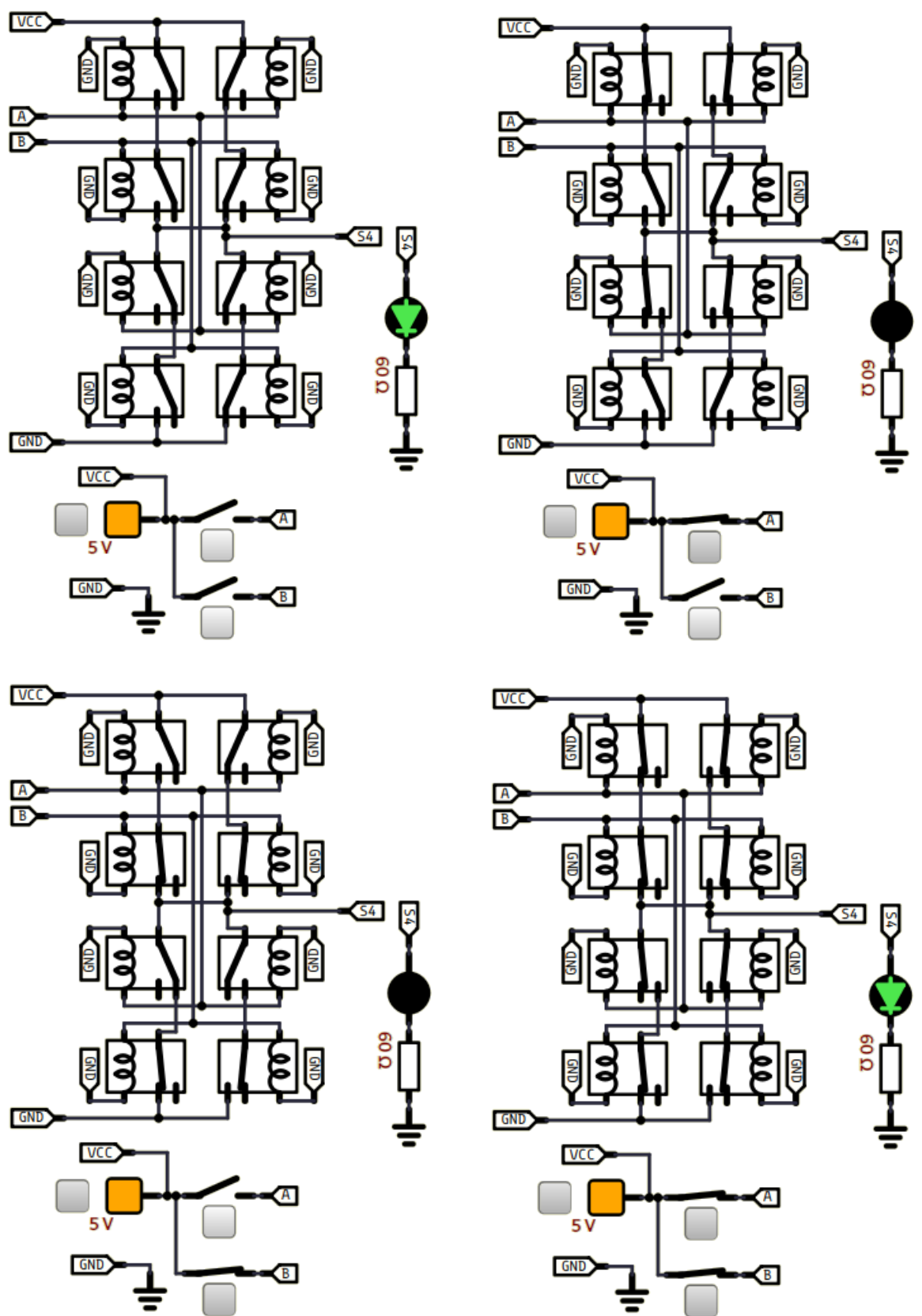
Transistors
(logisim)

Équivalences

XOR => $3 \times 4 = 12$ TransistorsXOR => $3 \times 4 = 12$ TransistorsXOR => $3 \times 4 + 2 \times 2 = 16$ TransistorsXOR => $3 \times 4 + 2 = 14$ Transistors

(simulide)



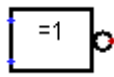


1-7) NON-OU-EXCLUSIF (XNOR) :

Américain

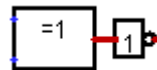


Europe



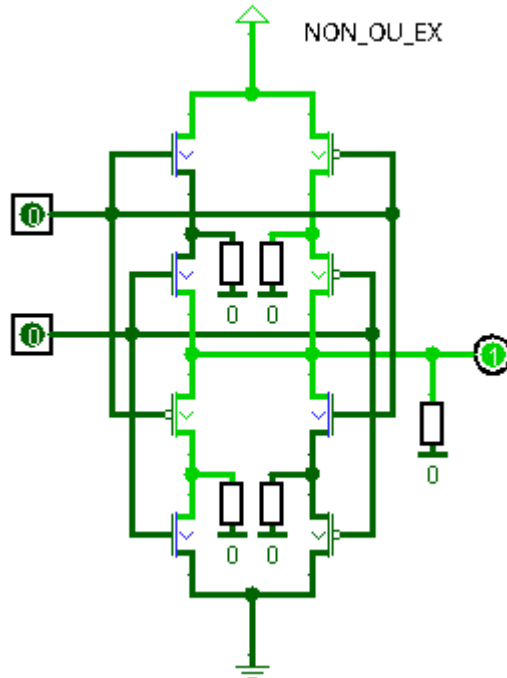
A	B	S
0	0	1
0	1	0
1	0	0
1	1	1

Équivalence

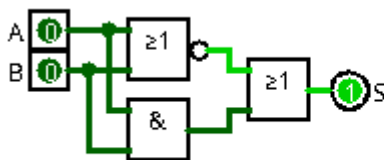
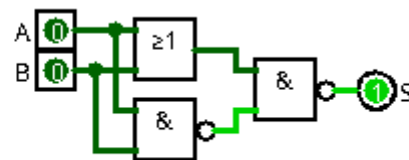
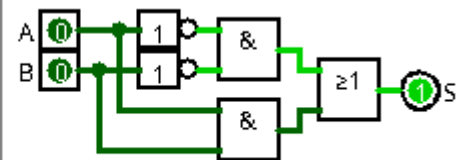


Transistors

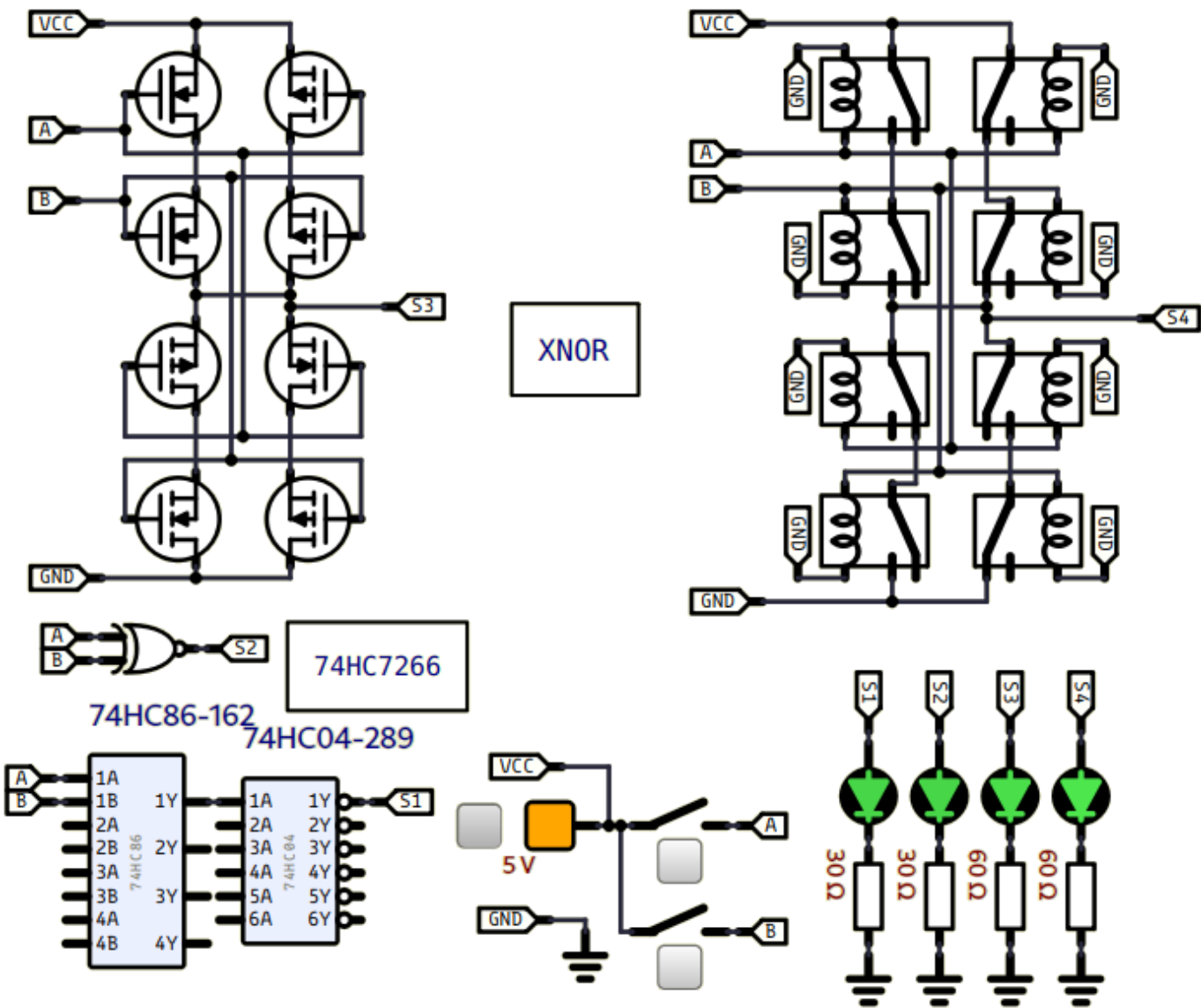
(logisim)

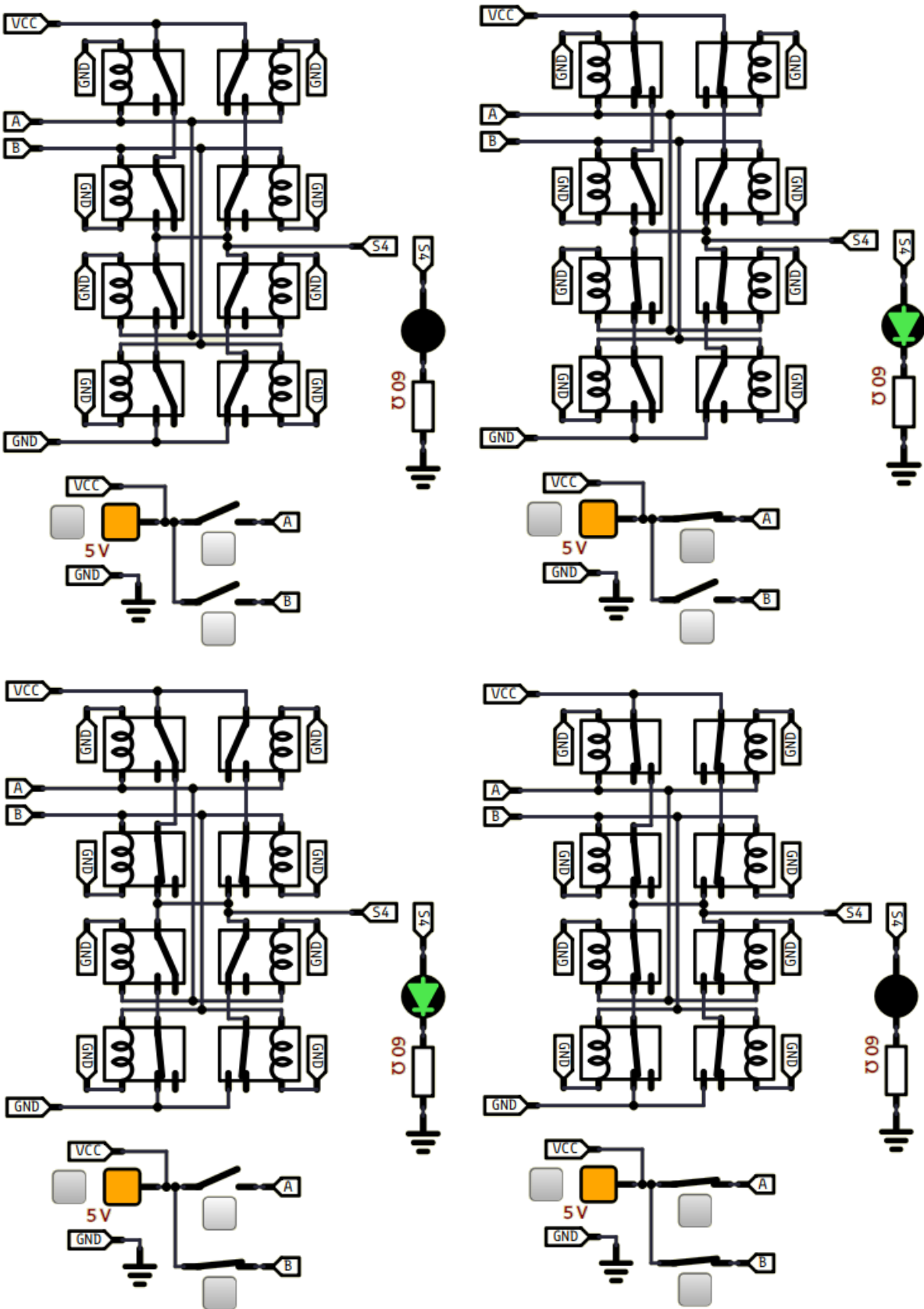
8 Transistors
NON_OU_EX

Équivalences

XOR => $3 \times 4 = 12$ TransistorsXOR => $3 \times 4 = 12$ TransistorsXOR => $3 \times 4 + 2 \times 2 = 16$ Transistors

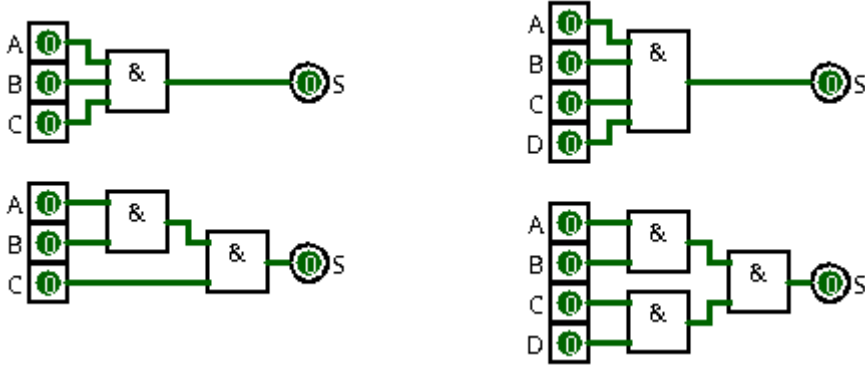
(simulide)





2) Portes logiques comportant plus de 2 entrées :

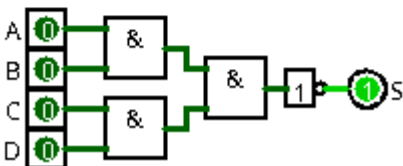
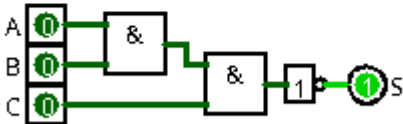
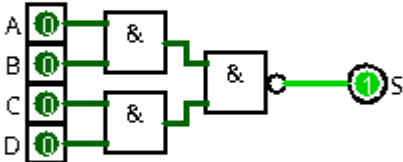
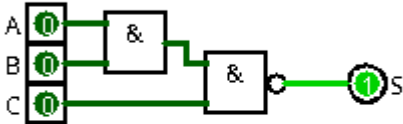
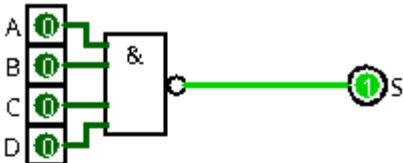
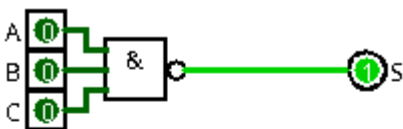
2-1) ET (AND) [3 à 4 entrées] :



A	B	C	S
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

A	B	C	D	S
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

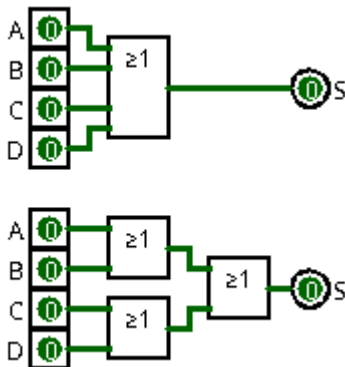
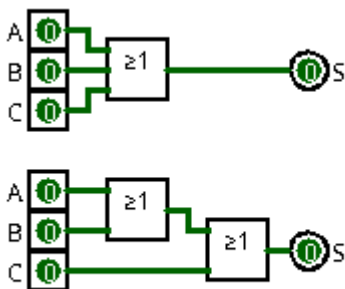
2-2) NON-ET (NAND) [3 à 4 entrées] :



A	B	C	S
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

A	B	C	D	S
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

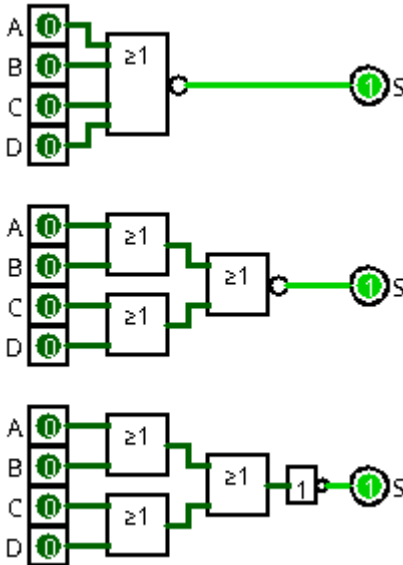
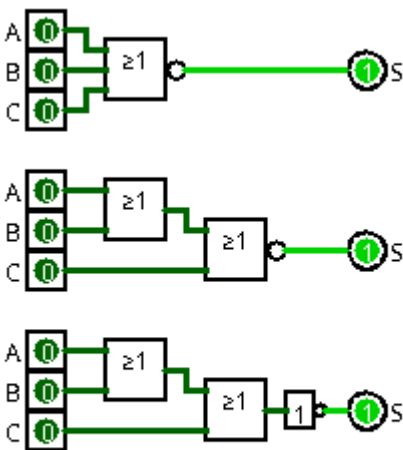
2-3) OU (OR) [3 à 4 entrées] :



A	B	C	S
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

A	B	C	D	S
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

2-4) NON-OU (NOR) [3 à 4 entrées] :

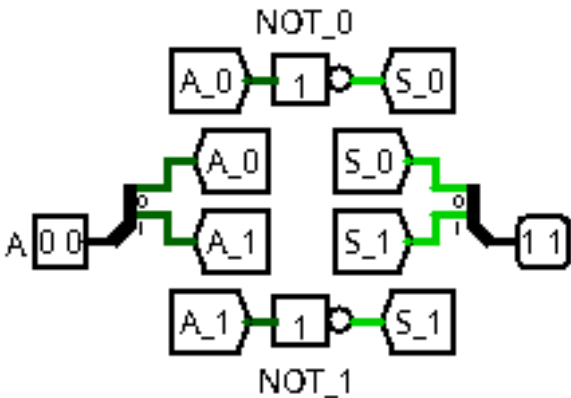
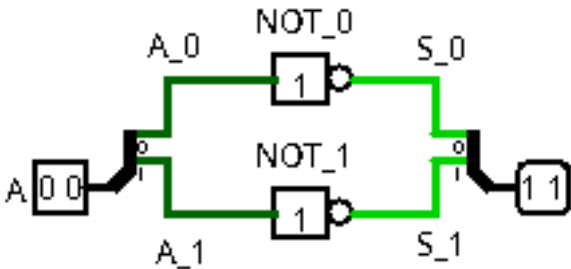


A	B	C	S
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

A	B	C	D	S
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

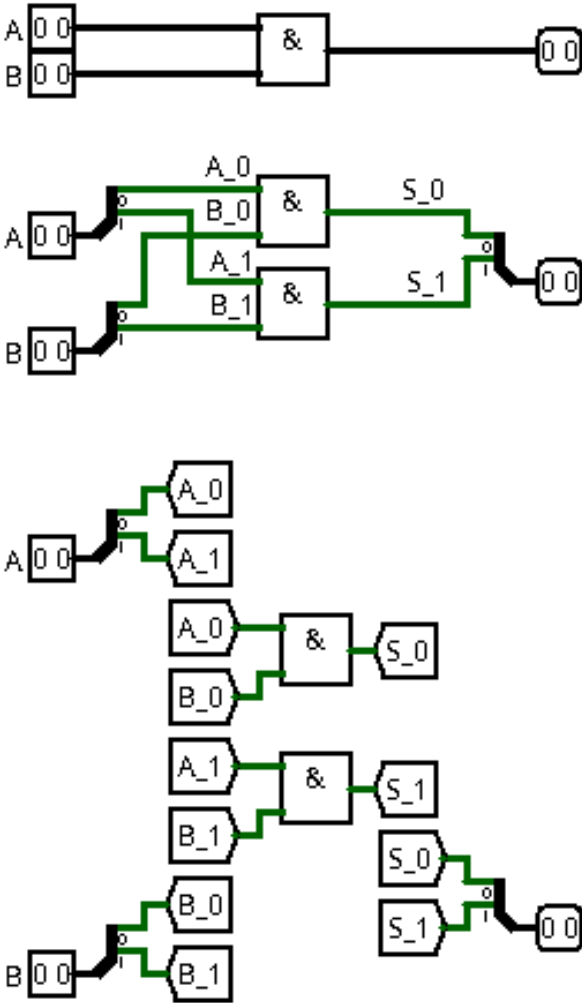
3) Portes logiques avec bus de données :

3-1) NON (NOT) {inverseur} :



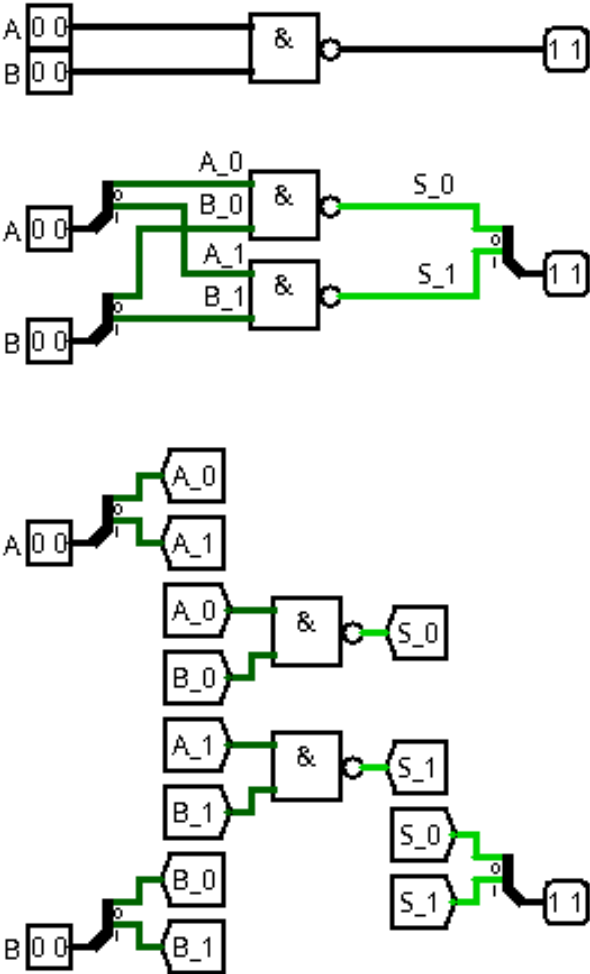
A		S	
0	0	1	1
0	1	1	0
1	0	0	1
1	1	0	0

3-2) ET (AND) :



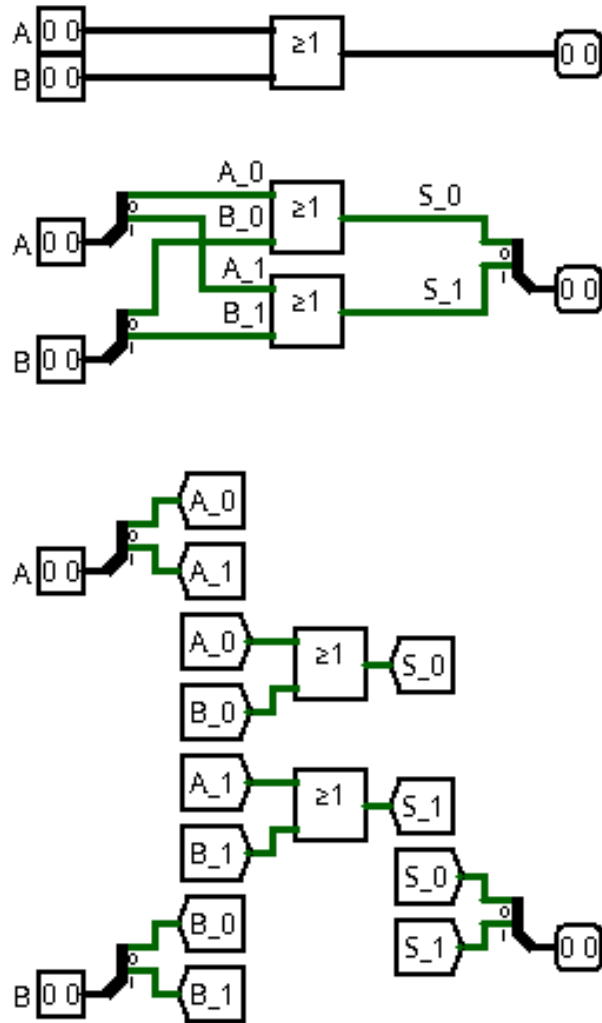
A		B		S	
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	0	0	0
0	0	1	1	0	0
0	1	0	0	0	0
0	1	0	1	0	1
0	1	1	0	0	0
0	1	1	1	0	1
1	0	0	0	0	0
1	0	0	1	0	0
1	0	1	0	1	0
1	0	1	1	1	0
1	1	0	0	0	0
1	1	0	1	0	1
1	1	1	0	1	0
1	1	1	1	1	1

1-3) NON-ET (NAND) :



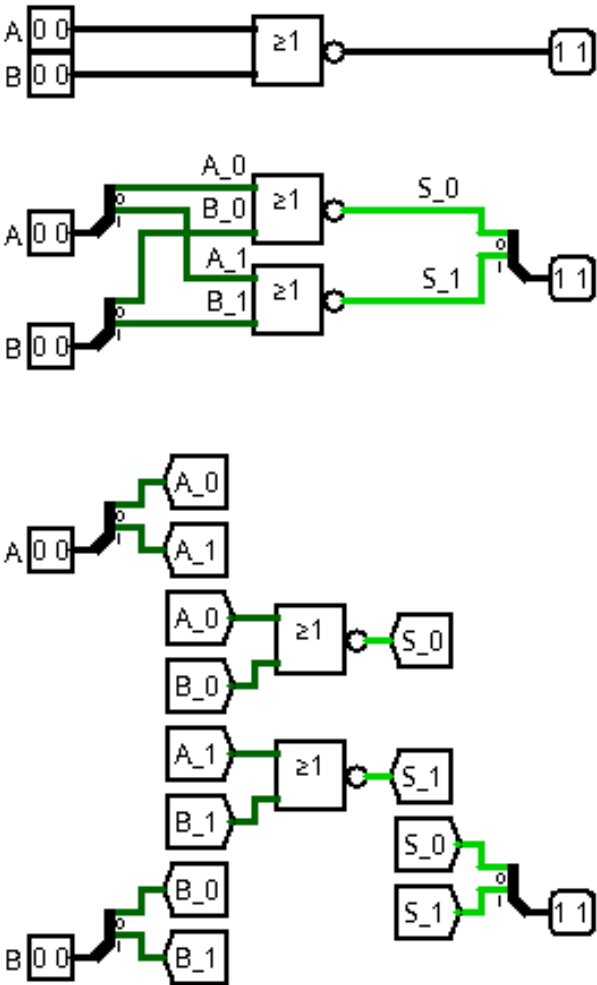
A		B		S	
0	0	0	0	1	1
0	0	0	1	1	1
0	0	1	0	1	1
0	0	1	1	1	1
0	1	0	0	1	1
0	1	0	1	1	0
0	1	1	0	1	1
0	1	1	1	1	0
1	0	0	0	1	1
1	0	0	1	1	1
1	0	1	0	0	1
1	0	1	1	0	1
1	1	0	0	1	1
1	1	0	1	1	0
1	1	1	0	0	1
1	1	1	1	0	0

3-4) OU (OR) :



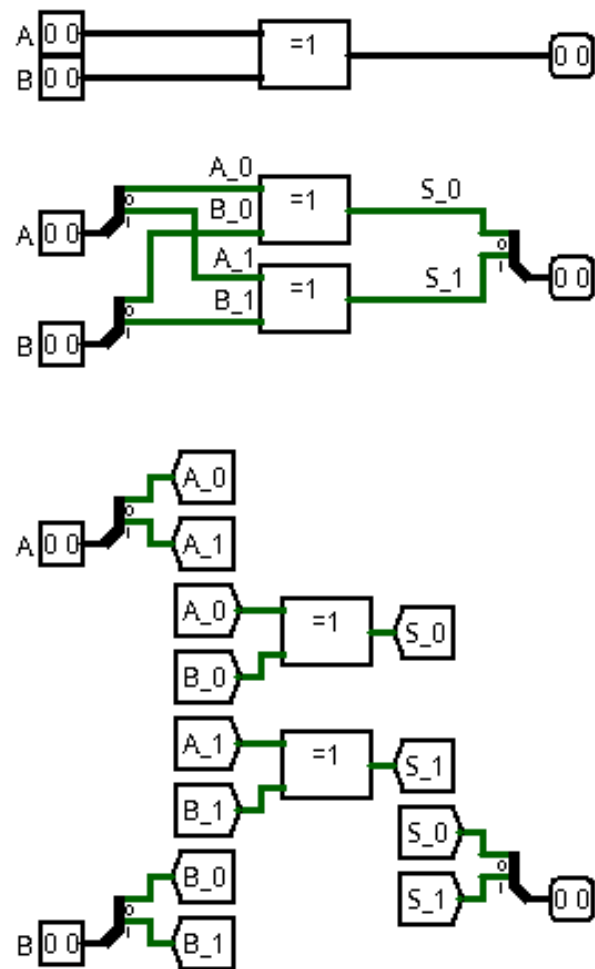
A		B		S	
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	1
0	1	0	0	0	1
0	1	0	1	0	1
0	1	1	0	1	1
0	1	1	1	1	1
1	0	0	0	1	0
1	0	0	1	1	1
1	0	1	0	1	0
1	0	1	1	1	1
1	1	0	0	1	1
1	1	0	1	1	1
1	1	1	0	1	1
1	1	1	1	1	1

3-5) NON-OU (NOR) :



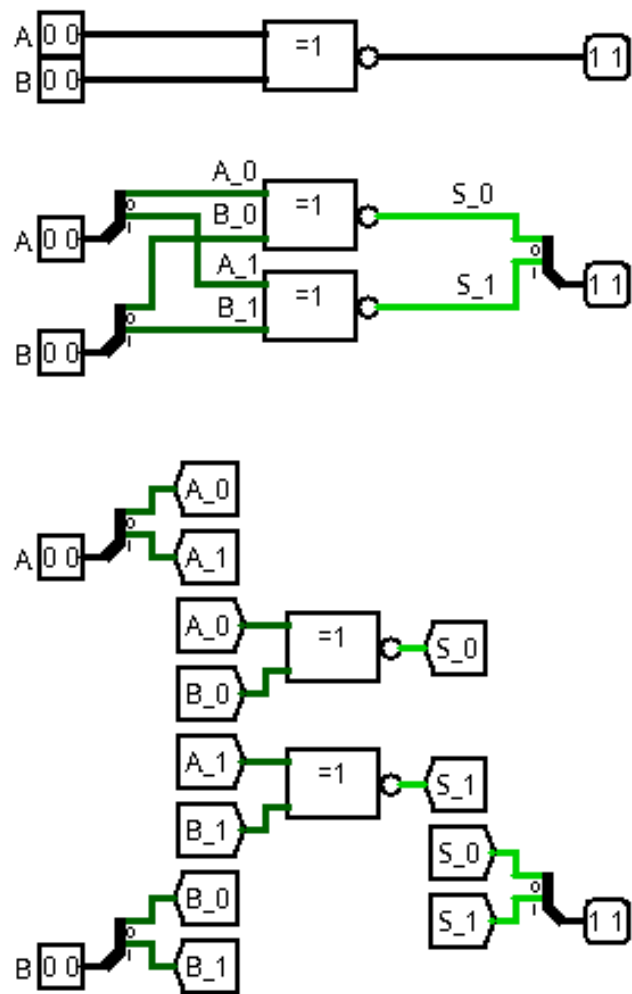
A		B		S	
0	0	0	0	1	1
0	0	0	1	1	0
0	0	1	0	0	1
0	0	1	1	0	0
0	1	0	0	1	0
0	1	0	1	1	0
0	1	1	0	0	0
0	1	1	1	0	0
1	0	0	0	0	1
1	0	0	1	0	0
1	0	1	0	0	1
1	0	1	1	0	0
1	1	0	0	0	0
1	1	0	1	0	0
1	1	1	0	0	0
1	1	1	1	0	0

3-6) OU-EXCLUSIF (XOR) :



A		B		S	
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	1
0	1	0	0	0	1
0	1	0	1	0	0
0	1	1	0	1	1
0	1	1	1	1	0
1	0	0	0	1	0
1	0	0	1	1	1
1	0	1	0	0	0
1	0	1	1	0	1
1	1	0	0	1	1
1	1	0	1	1	0
1	1	1	0	0	1
1	1	1	1	0	0

3-7) NON-OU-EXCLUSIF (XNOR) :



A		B		S	
0	0	0	0	1	1
0	0	0	1	1	0
0	0	1	0	0	1
0	0	1	1	0	0
0	1	0	0	1	0
0	1	0	1	1	1
0	1	1	0	0	0
0	1	1	1	0	1
1	0	0	0	0	1
1	0	0	1	0	0
1	0	1	0	1	1
1	0	1	1	1	0
1	1	0	0	0	0
1	1	0	1	0	1
1	1	1	0	1	0
1	1	1	1	1	1